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A novel deep learning approach for facial emotion recognition: application to detecting emotional responses in elderly individuals with Alzheimer’s disease

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Abstract

Facial expressions play a critical role in human communication by effectively conveying a wide range of emotions. Recent advancements in artificial intelligence and computer vision have made deep neural networks powerful tools for recognizing facial emotions. In a prior study, we introduced EmoNeXt, an innovative deep learning framework for facial expression recognition based on a modified ConvNeXt architecture, incorporating several significant enhancements. Initially, we evaluated its performance on the FER2013 dataset. This paper extends the evaluation of EmoNeXt to additional well-known benchmark datasets, including AffectNet and CK+, to further validate its robustness and generalizability. EmoNeXt integrates enhancements such as Spatial Transformer Networks, which help the model focus on areas of the face rich in expressive features, Squeeze-and-Excitation blocks, which improve the model’s ability to understand dependencies between different channels, and a self-attention regularization term that encourages the model to produce compact and effective feature vectors. Furthermore, this paper presents an extensive ablation study conducted on the three datasets, analyzing and confirming the contribution of each added module of our EmoNeXt model. Finally, we explore EmoNeXt’s application in recognizing emotions in

elderly individuals with Alzheimer’s disease, addressing the urgent need for accurate emotion recognition to improve patient care.

Keywords: Facial Expression Recognition (FER), Deep-learning, Convolutional neural network, Emotion detection, Emotional Artificial Intelligence, Alzheimer’s disease

1 Introduction

Facial expressions serve as a powerful tool in non-verbal communication, allowing individuals to convey a broad spectrum of emotions and intentions. As Albert Mehrabian famously highlighted, facial expressions outweigh words in face-to-face interactions, with only 7% of communication relying on verbal language [1]. This underscores the pivotal role facial expressions play in human communication, influencing perception and understanding.

Given this significance, modeling human perception of facial expressions has gained considerable interest. Over the past two decades, research in computer vision and artificial intelligence has focused extensively on the automatic recognition of facial expressions. Facial Expression Recognition (FER) has attracted substantial attention across various domains, including security, healthcare, driver fatigue surveillance, and human-machine interaction [2–4].

In recent years, numerous conventional approaches to FER have emerged, using classical descriptors to explicitly extract features from facial data. These approaches can be broadly categorized into two groups: geometric methods and appearance-based methods. Geometric features capture the shape, position, and interrelations of facial components during expressions [5, 6], while appearance-based features focus on variations in facial appearance, such as wrinkles and furrows, and can be extracted from the entire face or specific regions [3, 7, 8]. To classify these features, which encompass both geometric and texture-based characteristics, various classifiers have been employed, including Support Vector Machines (SVM) [9], k-Nearest Neighbors (kNN) [10], and neural networks such as Multi-Layer Perceptron (MLP).

Conventional FER approaches typically involve a two-step process: first, analyzing and defining facial features, and then utilizing these features to determine the emotion. However, performing these steps separately often leads to sub-optimal performance, especially when dealing with complex datasets containing numerous sources of variability. Consequently, combining these two steps can lead to better recognition performance.

In recent decades, deep neural networks have showcased remarkable effectiveness in automatic recognition tasks, making them well-suited for automatic FER [11, 12]. Deep learning facilitates the learning of discriminative feature representations for automatic FER by constructing a hierarchical architecture comprising multiple non-linear transformations, leveraging various types of neural networks such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs). These networks are complemented by a classification layer for the classification task. Consequently, the learning parameters of the classifier are determined concurrently with the learning of

the feature representations. This integration of feature extraction and classification directly from raw data significantly reduces reliance on models based on face geometry and other preprocessing techniques.

In this paper, we expand on the foundational research introduced in [13] and further evaluate our deep learning framework for FER, which we have named EmoNeXt. Originally detailed in [13], EmoNeXt was assessed solely using the FER2013 dataset [14], where it demonstrated superior performance over existing models. EmoNeXt has been enhanced by integrating a Spatial Transformer Network (STN) [15], enabling the learning and application of spatial transformations to input images, as well as Squeeze-and-Excitation (SE) blocks [16] for adaptive recalibration of channel-wise features. Additionally, we introduced a self-attention regularization term along with the classical cross-entropy loss [17] to encourage the generation of compact features by the model.

In this extended study, we broaden our evaluation to include two additional prominent and distinguished datasets: AffectNet [18], the largest FER dataset collected from real-world scenarios, and CK+ [19], a well-respected posed facial dataset. Additionally, we conduct an extensive ablation study to assess the impact of each enhancement integrated into EmoNeXt. This comprehensive evaluation not only analyzes the specific contributions of each added module but also examines their collective effects on emotion recognition performance. The results are crucial in deepening our understanding of the underlying mechanisms of EmoNeXt and identifying the most influential aspects of our approach. In our evaluation, we compare EmoNeXt’s performance against the latest and most effective facial emotion recognition models, showing that EmoNeXt consistently surpasses both the original ConvNeXt network and other state-of-the-art deep learning models across multiple FER datasets, including FER2013, CK+, and AffectNet.

Finally, to further highlight the practical implications of our work, we explore an important application of facial emotion recognition in the healthcare sector, particularly for elderly individuals suffering from Alzheimer’s disease (AD). AD patients often experience significant impairments in verbal and non-verbal communication, making it challenging to accurately gauge their emotional states. This difficulty arises due to the neurodegenerative effects of AD, which limit the range of emotions these individuals can express [20–23]. Accurate detection of emotions such as neutrality, anger, sadness, and joy in AD patients can significantly enhance their care and improve their quality of life. However, to our knowledge, no prior studies exist that classify the emotions of AD patients using facial expressions to improve their care. In this study, we demonstrate the potential of EmoNeXt in detecting emotions in elderly individuals with Alzheimer’s disease, highlighting its applicability and effectiveness in real-world healthcare scenarios.

The paper is structured as follows: Section 2 provides a review of relevant literature in the field of FER, with a particular emphasis on deep learning models evaluated using benchmark datasets such as FER2013, CK+, and AffectNet. Section 3 elaborates on the architecture of our proposed method, EmoNeXt. Section 4 presents a series of analyses and visualizations demonstrating the superiority of our model, including its application to detecting emotions in elderly AD patients, thus highlighting its real-world applicability. Section 5 concludes the paper by summarizing our findings, and

Section 6 outlines various ideas and methodologies planned for future enhancements to our proposed model.

2 Related Work

In this section, we briefly review recent research on facial emotion recognition, focusing on models evaluated on the widely recognized FER2013, CK+, and AffectNet datasets. These datasets are among the most extensively used in the field of FER, serving as benchmarks for evaluating algorithmic performance and generalization. We emphasize deep learning approaches, as our paper focuses on automated methods in this category, omitting classical methods using handcrafted descriptors combined with machine learning classifiers.

Given the significant advancements and widespread adoption of deep learning in computer vision, particularly in tasks like image classification, numerous studies have proposed various deep learning methodologies for automated FER tasks. These studies typically use benchmark datasets with the common goal of achieving the highest possible classification accuracy.

In their research [24], the authors introduced the Deep Attentive Center Loss (DACL) method to enhance discrimination by adaptively selecting a subset of significant feature elements. DACL integrates an attention mechanism that estimates attention weights based on the intermediate spatial feature maps in a CNN, providing context to identify feature importance. These estimated weights facilitate the sparse formulation of center loss, promoting intra-class compactness and inter-class separation for relevant information in the embedding space. Empirical results showcased the superiority of DACL over baseline methods (softmax loss and center loss) and other state-of-the-art techniques on RAF-DB and AffectNet datasets.

Another notable contribution comes from [25], where a novel channel self-attention module named Local multi-Head Channel (LHC) was introduced. This LHC module, tailored for computer vision tasks, particularly emphasizes facial expression recognition. Unlike global attention mechanisms, LHC effectively focuses on crucial facial features, thereby mitigating their limitations. Impressively, LHC seamlessly integrates into existing CNN architectures. The proposed LHC-Net not only achieved a new state-of-the-art performance on the FER2013 dataset but also demonstrated significantly lower computational complexity and architectural impact compared to other contemporary methods.

In their recent study [26], Han et al. introduced a novel symmetric approach for extracting inter-class and intra-class diversity features, proposing a triple-structure network model trained with a multi-branch loss function. This innovative network consists of three distinct structures: a global branch network, an attention mechanism branch network, and a diversified feature learning branch network. Leveraging MobileNet V1 [27] as its backbone, renowned for its lightweight design and high recognition accuracy, the proposed model was evaluated on the KDEF, MMI, and CK+ datasets, achieving commendable classification accuracies. Through meticulous ablation experiments and insightful visualizations, the authors provided compelling evidence supporting the efficacy and rationality of their triple-structure network model.

In their work, Fard et al. introduced the Adaptive Correlation (Ad-Corre) Loss, designed to guide neural networks in generating embedded feature vectors characterized by high within-class correlation and low between-class correlation [28]. Comprising three essential components—Feature Discriminator (FD), Mean Discriminator (MD), and Embedding Discriminator (ED)—this loss function was seamlessly integrated with the Cross-Entropy Loss and applied to train two prominent backbone models: Xception [29] and ResNet50 [30]. The authors demonstrated the effectiveness of the Ad-Corre Loss across various deep learning architectures, showcasing remarkable improvements in discriminative power and classification accuracy. Notably, their approach achieved outstanding recognition accuracies on benchmark datasets such as AffectNet, RAF-DB, and FER2013.

Another noteworthy contribution comes from [31], where Vignesh et al. introduced the Segmentation VGG-19 model, aiming to enhance facial expression recognition tasks by incorporating segmentation blocks inspired by U-Net [32] into the VGG-19 architecture [33]. These segmentation blocks function as attention mechanisms, enabling the model to focus on salient facial features. By interleaving these blocks within VGG-19 layers, the proposed model effectively enhances feature extraction, resulting in superior performance compared to both the baseline VGG-19 network and other state-of-the-art classification models, as demonstrated on the FER2013 dataset under identical experimental conditions.

In a recent study by Meena et al. [34], an efficient end-to-end approach for sentiment analysis was introduced, leveraging the well-known Inception-V3 network [35]. Through systematic fine-tuning of the Inception-V3 network using a pipeline training technique, the authors significantly improved its performance for sentiment analysis tasks. Evaluation of the proposed approach on established facial image datasets, including FER2013, JAFFE, and CK+, showcased exceptional accuracies, demonstrating its effectiveness in recognizing individuals’ emotions.

In their recent publication [36], Zhang et al. presented the Dual-Direction Attention Mixed Feature Network (DDAMFN), a novel architecture tailored for FER, characterized by robustness and lightweight design. The DDAMFN comprises two core components: the Mixed Feature Network (MFN) for resilient feature extraction using mixed-size kernels, and the Dual-Direction Attention Network (DDAN) for generating attention maps in two orientations. This dual-directional attention mechanism enables the model to effectively capture long-range dependencies in facial expressions. Experimental evaluations conducted on diverse public datasets, including AffectNet, RAF-DB, and FERPlus, unequivocally demonstrated the superiority of the DDAMFN over existing models.

In a recent investigation by Punuri et al. [37], an efficient scheme named EfficientNet-XGBoost was introduced for facial emotion recognition. This scheme combines a pre-trained EfficientNet architecture with fully connected layers, an XGBoost classifier, and customized fine-tuning of parameters. Thorough validation on benchmark datasets including CK+, KDEF, JAFFE, and FER2013 demonstrated impressive accuracies, highlighting the novelty and efficacy of the proposed method.

Wen et al. [38] introduced the Distract your Attention Network (DAN), comprising the FCN (Feature Clustering Network), MAN (Multi-head Attention Network),

and AFN (Attention Fusion Network). The FCN extracts robust features using a large-margin learning objective, while the MAN employs multiple attention heads to focus on different facial regions simultaneously. The AFN redirects these attentions to diverse locations before merging them into a comprehensive representation. Experimental results on AffectNet, RAF-DB, and SFEW 2.0 consistently showcased state-of-the-art performance in facial expression recognition.

Lu et al. proposed a Multi-channel Attention Network for expression analysis in natural scenes [39]. This network integrates ResNet18 [30] and ConvNeXt concepts [40] for feature extraction and employs adaptive feature fusion and channel weighting. The attention module refines the spatial attention mechanism to prioritize feature information acquisition. Experimental results demonstrated superior recognition efficiency on the FER2013 dataset under uncontrolled conditions and notable accuracies on the Oulu_CASIA and RAF-DB datasets, underscoring the effectiveness of the proposed method.

In their recent study, Zheng et al. introduced the POSTER (Pyramid crOSS-fuSion Transformer) network, targeting three critical challenges in facial expression recognition: inter-class similarity, intra-class discrepancy, and scale sensitivity [41]. The POSTER leverages a transformer-based cross-fusion method to effectively integrate facial landmark and image features, ensuring optimal attention to salient facial regions. Additionally, its pyramid structure enhances scale invariance, resulting in unprecedented state-of-the-art performance on datasets such as RAF-DB, FERPlus and AffectNet.

In another recent work, Zhang et al. introduced a novel cross-fusion dual-attention network operating across spatial and channel dimensions [42]. This network comprises a cross-fusion grouped dual-attention mechanism, a C^2 activation function construction method, and a closed-loop operation integrating self-attention distillation with residual connections. Experimental evaluations on RAF-DB, FERPlus, and AffectNet datasets demonstrated its effectiveness in addressing FER tasks.

Given the breadth of research in the FER domain, it is impractical to discuss all related deep learning-based models within this section. To address this, the Results subsection (4.3) includes an extensive compilation of additional references to significant FER models. This compilation is supplemented by detailed tables that summarize the classification scores achieved by these state-of-the-art FER models on prominent datasets, including FER2013, CK+, and AffectNet. These tables provide a clear benchmark for comparing the performance of EmoNeXt with other leading approaches in the field.

3 Materials and Methods

This section explores the various techniques and components integrated into a comprehensive deep learning model designed specifically for facial emotion detection. The focus primarily revolves around three key aspects: preprocessing input images, feature extraction and classification, and loss function minimization.

3.1 Spatial Transformer Networks

Spatial Transformer Networks (STN) [15] extend the concept of differentiable attention to encompass multiple spatial transformations. By integrating a differentiable geometric transformation module into the network architecture, STN allows neural networks to learn and apply spatial transformations to input data. This capability is invaluable in FER, where performance is heavily influenced by scale, rotation, and translation variations.

The spatial transformer mechanism comprises three main components as illustrated in Figure 1:

1. **Localization Network:** This sub-network takes the input image and learns to predict the parameters of the spatial transformation. It usually consists of one or more convolutional layers followed by fully connected layers that output the transformation parameters. These parameters include the translation vector (t_x, t_y) , the scale factors (s_x, s_y) , and the rotation angle θ . The transformation parameters can be represented using a 3x2 matrix. This matrix is often referred to as the affine transformation matrix or the theta matrix and it's defined as:

$$A_\theta = \begin{bmatrix} s_x \times \cos(\theta) & -s_y \times \sin(\theta) & t_x \\ s_x \times \sin(\theta) & s_y \times \cos(\theta) & t_y \end{bmatrix}$$

2. **Grid Generator:** The grid generator takes the predicted affine matrix from the localization network and generates a set of sampling grid coordinates. These grid coordinates define the transformation to be applied to the input image. The grid can be seen as a set of control points that define how the input pixels should be transformed.
3. **Sampler:** The sampler module applies the spatial transformation defined by the grid to the input image. It performs differentiable sampling by bilinear interpolation, which allows gradients to be backpropagated through the transformation. The sampler module takes the input image and the grid coordinates as inputs and produces the transformed image as the output.

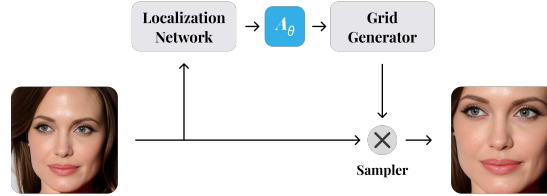


Fig. 1 The architecture of a spatial transformer module.

During training, the STN is trained end-to-end with the rest of the network. The parameters of the localization network are learned through backpropagation, and the gradients flow through the sampler module to update both the localization network and the rest of the network's parameters. The key advantage of STN is their ability

to learn spatial transformations automatically as part of the neural network training process.

3.2 ConvNeXt

Introduced in 2022, ConvNeXt [40] is a pure convolutional model that draws inspiration from Vision Transformers [43]. It was designed to compete with state-of-the-art Vision Transformers while retaining the simplicity and efficiency of CNNs. The authors gradually "modernized" the architecture from a standard ResNet [30] to a hierarchical vision Transformer (such as Swin-T [44]) to explore the impact of design decisions in Transformers on CNNs performance. Along the way, they discovered key components contributing to the performance difference.

Constructed entirely from standard CNN modules, ConvNeXt demonstrated favorable competition with Transformers in terms of accuracy, scalability, and robustness across major vision benchmarks. These ConvNeXts maintained the efficiency of CNNs and were fully convolutional, making them simple to implement for both training and inference.

The architecture of ConvNeXt, as illustrated in Figure 4, follows a multistage design approach that incorporates varying feature map resolutions for each stage. Two main design considerations are the stage-compute-ratio (SCR), represented by the number of blocks per stage, and the structure of the stem cell. ConvNeXt consists of four stages, with SCR values set to (3, 3, 9, 3).

One notable distinction between ConvNeXt and ResNet (as well as other CNNs) is the utilization of a patchify layer in ConvNeXt. This layer is implemented through a 4×4 convolutional layer with a stride of 4. In contrast, ResNet and general CNNs employ a stem cell composed of a 7×7 convolutional layer with a stride of 2, followed by a max-pooling operation.

The ConvNeXt block, depicted in Figure 2, incorporates several modifications to improve performance and reduce floating-point operations (FLOPs). It employs larger kernel-sized convolutions and depthwise convolutions, widens the network to 96 channels, and adopts an inverted bottleneck design. In addition, it replaces the ReLU [45] activation function with GELU [46] and replaces BatchNorm (BN) [47] with Layer Normalization (LN)[48]. These changes contribute to a slightly better performance, with GELU and LN commonly utilized in advanced Transformers.

The ConvNeXt block consists of a 7×7 depthwise convolution, two 1×1 layers, and a nonlinear GELU activation function. Prior to the Conv 1×1 layer, Layer Normalization (LayerNorm) is applied. The depthwise convolution, a form of grouped convolution where the number of groups and channels are equal, is analogous to the per-channel weighted sum operation in self-attention mechanisms, facilitating information mixing in the spatial dimension.

Additionally, the ConvNeXt architecture incorporates a separate downsampling layer between stages. This downsampling is achieved using 2×2 Conv layers with a stride of 2.

The authors of ConvNeXt have developed multiple versions, distinguished by variations in the number of channels (C) and blocks (B) within each stage. Here are the configurations:

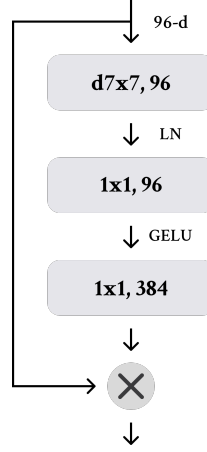


Fig. 2 The ConvNeXt block.

Tiny	$C = (96, 192, 384, 768)$	$B = (3, 3, 9, 3)$
Small	$C = (96, 192, 384, 768)$	$B = (3, 3, 27, 3)$
Base	$C = (128, 256, 512, 1024)$	$B = (3, 3, 27, 3)$
Large	$C = (192, 384, 768, 1536)$	$B = (3, 3, 27, 3)$
XLarge	$C = (256, 512, 1024, 2048)$	$B = (3, 3, 27, 3)$

3.3 Squeeze-and-Excitation Blocks

Squeeze-and-Excitation (SE) [16] is a powerful technique used in deep learning models to enhance the representational power of CNN models. It introduces a mechanism that allows the network to adaptively recalibrate the channel-wise features, improving the model’s discriminative capabilities.

The motivation behind the SE block stems from the observation that different channels in a CNN capture different levels of semantic information. Some channels may be more informative and contribute more significantly to the final representation, while others may be less informative or even redundant. The SE block aims to model these channel-wise dependencies and selectively amplify informative channels while suppressing less important ones.

The SE block consists of two key operations: squeeze and excitation (Figure 3) :

1. Squeeze operation: The squeeze operation captures the global distribution of channel-wise feature responses. It reduces the spatial dimensions of the feature maps, typically by performing global average pooling (GAP). The global average pooling operation computes the average value for each channel across the spatial dimensions, resulting in a channel-wise feature descriptor.

2. **Excitation operation:** The excitation operation models the interdependencies among the channel-wise descriptors obtained from the squeeze operation. It captures the relationships between channels and learns channel-wise weights that represent the importance of each channel. The excitation operation usually consists of two fully connected (FC) layers with non-linear activations in between. The first FC layer reduces the dimensionality of the descriptors, while the second FC layer expands it back to the original channel dimension. Non-linear activation functions such as the rectified linear unit (ReLU) or sigmoid are applied to introduce non-linearity into the excitation process.

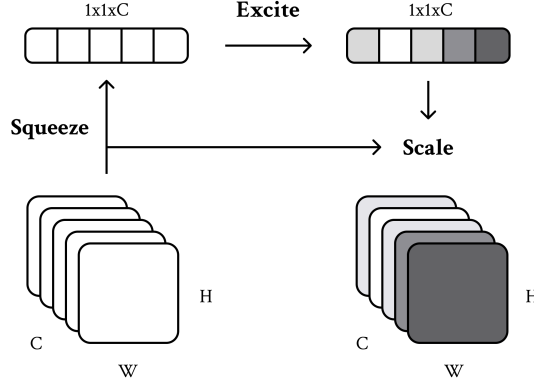


Fig. 3 The architecture of the Squeeze-and-Excitation block.

The output of the excitation operation is a set of channel-wise scaling factors or attention weights that capture the importance of each channel. These weights are then used to recalibrate the feature maps produced by the preceding convolutional layer. Specifically, each channel’s feature map is multiplied element-wise with its corresponding attention weight, allowing the network to emphasize informative channels and suppress less relevant ones.

3.4 EmoNeXt Final Architecture

The EmoNeXt architecture begins with the inclusion of an STN at the beginning of the network. The STN enables the model to handle variations in scale, rotation, and translation by learning and applying spatial transformations to facial images.

After passing through the STN, the inputs are processed by ConvNeXt’s patchify module. This module downsamples the input image using a non-overlapping convolution with a kernel size of 4. The downsampling reduces the dimensionality of the input and efficiently captures relevant features.

The downscaled inputs then go through the ConvNeXt stages. Each stage is followed by an SE block to recalibrate the feature map before proceeding to the next stage. This recalibration enhances the model’s ability to extract discriminative facial

features for accurate emotion recognition. The overall architecture is illustrated in Figure 4.

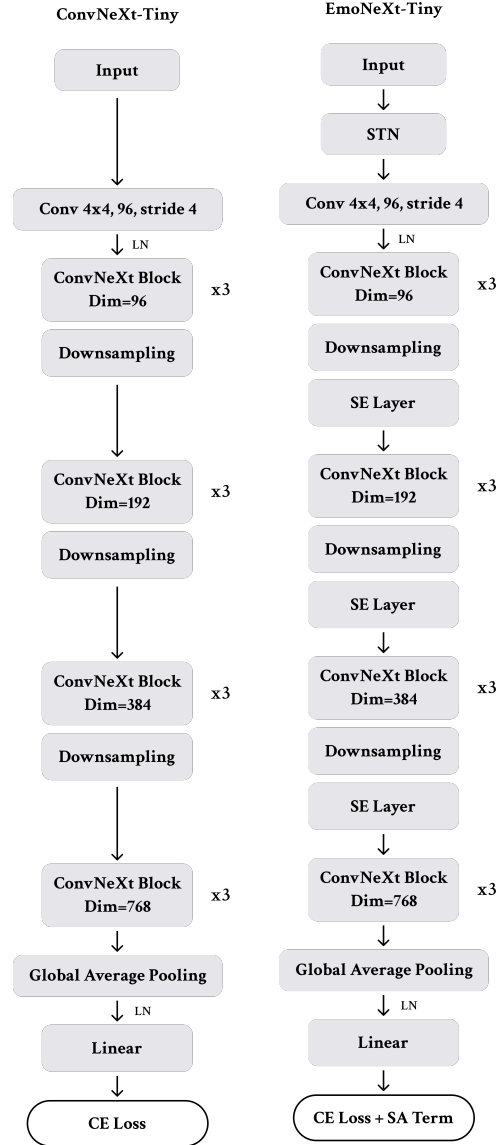


Fig. 4 Architecture designs for ConvNeXt and EmoNeXt.

By leveraging these techniques, our EmoNeXt model achieves robust and accurate facial emotion detection, effectively handling variations in facial expressions and improving overall performance.

3.5 Self-Attention Regularization Term

Self-attention is a powerful mechanism employed in various domains, including natural language processing, image analysis, and feature extraction [49]. It allows models to assign importance to different parts of the input sequence, effectively capturing dependencies and relationships.

One popular method is dot product self-attention, known for its simplicity and effectiveness. This approach calculates the relevance and importance of the feature vector by computing the dot product similarity between pairs of its elements. The resulting attention scores are then used to weight the feature vector, generating an attended representation that emphasizes the most significant information.

Mathematically, the dot product self-attention weights are computed as follows:

$$W(\mathbf{Q}, \mathbf{K}) = \text{softmax} \left(\frac{\mathbf{Q} \cdot \mathbf{K}^\top}{\sqrt{d}} \right) \quad (1)$$

Here, $\mathbf{Q}$ and $\mathbf{K}$ represent queries and keys of dimension d . Queries ($\mathbf{Q}$) and keys ($\mathbf{K}$) are computed from a feature vector using a linear layer. The dot product between $\mathbf{Q}$ and $\mathbf{K}^\top$ computes pairwise similarities, and scaling by $\sqrt{d}$ prevents gradient problems. The softmax operation normalizes the attention scores.

For optimizing the model’s performance, selecting an appropriate loss function is crucial. In multiclass classification tasks, Cross-Entropy Loss [17] is the standard choice. However, we introduce a novel component called the self-attention (SA) regularization term.

The SA regularization term aims to balance the importance between generated features by minimizing attention weights. It is defined as follows:

$$\mathcal{L}(W) = \frac{1}{N} \sum_{i=1}^N (W_i - \bar{W})^2 \quad (2)$$

Here, W represents the self-attention weights, $\bar{W}$ is the mean attention weight, and N is the length of the feature vector. Minimizing this term encourages the attention weights to be closer to the mean value, promoting balanced importance across the features.

This SA regularization term is added to the overall loss function of the model, in addition to the Cross-Entropy Loss. This incorporation helps during training by encouraging the model to produce more balanced attention weights and improve generalization.

The final loss function, combining the Cross-Entropy Loss and the SA regularization term, is:

$$\mathcal{L}_{\text{final}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \mathcal{L}_{\text{SA}} \quad (3)$$

Here, $\mathcal{L}_{\text{CE}}$ is the Cross-Entropy Loss, $\mathcal{L}_{\text{SA}}$ is the SA regularization term, and λ is a hyperparameter that controls the trade-off between the two terms.

4 Experimental results

4.1 Datasets

4.1.1 FER2013

Initially introduced at the ICML 2013 Challenges in Representation Learning [14], the FER2013 dataset comprises 35,887 grayscale images, each measuring 48×48 pixels. This dataset is segmented into 28,709 training images, 3,589 validation images, and 3,589 testing images. The images are categorized into seven emotion classes, as detailed in Table 1. FER2013 is widely utilized in facial expression recognition tasks due to its extensive sample size. However, it presents challenges for deep learning models due to its significantly unbalanced class distribution. Figure 5 showcases a selection of images from FER2013.

4.1.2 CK+

The CK+ dataset [19] includes 981 images spanning seven distinct emotion classes: fear, disgust, sadness, happiness, neutrality, surprise, and anger (see Table 1). Each image is a 48×48 pixel grayscale photograph, as illustrated in Figure 5. The CK+ dataset is integral during data merging phases, ensuring balanced class distribution and normalization of feature variations. Images in CK+48, typically extracted from video frames, show minimal variability and are predominantly frontal, clearly depicting facial expressions. This distinguishes CK+ from other datasets like FER2013 and AffectNet due to its focused representation of clear facial cues.

4.1.3 AffectNet

AffectNet contains 287,651 training images and 4,000 manually annotated validation images [18] (see Table 1 and Figure 5). Notably, its test set is not publicly available. Research evaluations [18] primarily utilize the validation set, which includes 500 images per emotion, covering eight classes: anger, contempt, disgust, fear, happiness, neutral, sadness, and surprise. Given the typical exclusion of contempt from the standard seven-class model of facial expression tasks, several studies [24, 28, 50, 51] choose to focus on a modified subset of the validation set, excluding the images labeled as contempt. In our evaluations, we apply both 8-way and 7-way classification settings to ensure a thorough comparison with previous research [36, 38, 41, 52].

Table 1 FER2013, AffectNet and CK+ datasets summary

Dataset	Angry	Disgust	Fear	Happy	Sad	Surprise	Contempt	Neutral	Total
FER2013	4953	547	5121	8989	6077	4002	0	6198	35887
AffectNet	25382	4303	6878	134915	25959	14590	4250	75374	291651
CK+	135	177	75	207	84	249	54	0	981



Fig. 5 Images from the datasets: CK+, FER2013 and AffectNet.

4.2 Training

Recent studies have underscored the efficacy of contemporary training methodologies in significantly enhancing deep learning model performance. For our model training, we employ a variety of advanced strategies. We utilize the AdamW optimizer [53] with a learning rate of $1e-4$, combined with a cosine decay schedule to improve convergence. We also implement data augmentation techniques such as Random Cropping and Random Rotation, which expand the training dataset and boost the model’s generalization capabilities. To prevent overfitting, we apply regularization strategies like Stochastic Depth [54] and Label Smoothing [35], contributing to more robust and generalized models.

Further, to enhance performance while managing memory constraints, we incorporate the Exponential Moving Average (EMA) technique [55], known for mitigating overfitting in larger models. Additionally, we utilize Mixed Precision Training [56], which reduces memory usage by approximately 50% and accelerates training.

Moreover, we enhance our model’s performance by integrating weights from pre-trained ConvNeXt models using the ImageNet-22k dataset [57]. This dataset is renowned for its extensive collection of diverse images, enabling our model to benefit from a substantial amount of pre-learned knowledge. To align with the pretrained models dimensions, we resize our training images from the FER2013 and CK+ datasets to 224×224 pixels, conforming to industry standards and optimizing the use of pre-trained weights for improved model efficacy. As for the AffectNet dataset, its images are already resized to this dimension, eliminating the need for further preprocessing.

Training for both the EmoNeXt and ConvNeXt models, including ablation studies across five model sizes (Tiny, Small, Base, Large, and XLarge), was conducted using an Nvidia T4 GPU with 16GB of VRAM. The models were implemented in PyTorch version 2.0.1. The source code for our EmoNeXt architecture is available on GitHub at: <https://github.com/yelboudouri/EmoNeXt>.

4.3 Results

4.3.1 Ablation experiments

To evaluate the effectiveness of our proposed EmoNeXt model across various sizes—Tiny, Small, Base, Large, and XLarge—an extensive ablation study was conducted. This study examined the influence of the Spatial Transformer Network (STN), Squeeze-and-Excitation (SE) modules, and the Self-Attention (SA) Regularization Term on the FER2013, CK+, and AffectNet datasets, as presented in Table 2.

Our EmoNeXt model primarily consists of three branches: a global branch network (ConvNeXt), an attention branch network (SE), and a Spatial Transformer branch network (STN). To assess its effectiveness, we conducted comparisons with three configurations: ConvNeXt, ConvNeXt+STN, and ConvNeXt+STN+SE. In ConvNeXt, only the original branch network is utilized. ConvNeXt+STN incorporates the Spatial Transformer branch module into the original branch network, enabling spatial transformations. Finally, ConvNeXt+STN+SE includes attention branch modules, further enhancing feature extraction and refinement within the global branch network.

The Original ConvNeXt configuration (the first row of each size) denotes the absence of the STN and the SE modules, as well as the SA Regularization Term in the ConvNeXt architecture. In this setup, input images are processed through the stem layer, and subsequent feature maps are directly fed into the ConvNeXt blocks without any enhancement from STN. The inclusion of the STN module marginally improves performance for all sizes, with notable gains in spatial transformation capabilities. More substantial improvements are achieved with the addition of the SE modules, which further refines feature extraction by capturing channel relationships and highlighting informative features while suppressing noise.

The introduction of the SA Regularization Term yields significant performance improvements, achieving state-of-the-art results across all sizes. This regularization mechanism enhances the model’s ability to capture long-range spatial dependencies, enriching the representation of facial expressions and leading to better classification performance.

4.3.2 Comparison with Other Methods

Tables 3, 4, 5, and 6 present our results and compare them with several state-of-the-art methods on FER2013, CK+, AffectNet-7, and AffectNet-8, respectively.

Results on the FER2013 dataset.

We compare the five versions of our model, EmoNeXt, against recent state-of-the-art FER models on the FER2013 dataset, as shown in Table 3. The results indicate that the XLarge version of EmoNeXt achieves state-of-the-art performance on FER2013. Specifically, EmoNeXt-XLarge improves upon Segmentation VGG-19 (75.97) by 0.15, CNNs+BOVW (75.42) by 0.7, and LHC-Net (74.42) by 1.7. This highlights the superiority of EmoNeXt-XLarge on FER2013.

Table 3 also compares the Tiny, Small, Base, and Large versions of EmoNeXt. While EmoNeXt-XLarge demonstrates the highest performance, the other versions

Table 2 Ablation results on the FER2013, AffectNet-7, AffectNet-8, and CK+ datasets. Our combinations of ConvNeXt, Spatial Transformer Network, Squeeze-and-Excitation module, and the Self-Attention Regularization Term are compared with each version of the original ConvNeXt model. The best result on each dataset is highlighted in bold.

Model	FER2013	AffectNet-7	AffectNet-8	CK+
ConvNeXt-T	71.99	63.38	60.56	92.23
ConvNeXt-T + STN	72.46	64.29	60.67	93.98
ConvNeXt-T + STN + SE	72.67	64.54	61.05	94.42
EmoNeXt-T	73.34	65.55	61.36	95.1
ConvNeXt-S	72.34	64.03	60.63	95.43
ConvNeXt-S + STN	73.59	64.46	61.35	96.21
ConvNeXt-S + STN + SE	73.92	64.82	61.44	97.2
EmoNeXt-S	74.33	65.9	62.51	97.78
ConvNeXt-B	73.22	64.12	61.11	96.55
ConvNeXt-B + STN	73.77	64.52	61.36	97.78
ConvNeXt-B + STN + SE	74.56	64.94	61.78	97.87
EmoNeXt-B	74.91	66.21	62.94	98.44
ConvNeXt-L	73.46	64.45	61.99	96.87
ConvNeXt-L + STN	74.74	64.83	62.22	98.33
ConvNeXt-L + STN + SE	74.97	65.11	62.63	98.88
EmoNeXt-L	75.57	66.88	63.12	100
ConvNeXt-XL	74.15	65.96	62.67	98.03
ConvNeXt-XL + STN	74.75	66.28	63.02	99
ConvNeXt-XL + STN + SE	75.68	66.87	63.49	99.43
EmoNeXt-XL	76.12	67.46	64.13	100

also perform competitively. For example, EmoNeXt-Tiny achieves an accuracy of 73.34, surpassing models like ResNet50 (73.20) and VGG (73.28).

EmoNeXt-Small shows enhanced performance with an accuracy of 74.33, outperforming advanced architectures such as the Residual Masking Network (74.14) and LHC-Net (74.42). EmoNeXt-Base maintains strong performance with an accuracy of 74.91, exceeding the performance of LHC-NetC (74.28). EmoNeXt-Large achieves an accuracy of 75.57, outperforming the CNN+BOVW model (75.42).

Overall, the superior performance of EmoNeXt-XLarge, with an accuracy of 76.12, surpasses the previous best state-of-the-art models, firmly establishing EmoNeXt-XLarge as a highly effective model for FER.

Results on the CK+ dataset.

We evaluate our EmoNeXt models against recent state-of-the-art FER methods on the CK+ dataset. As shown in Table 4, the XLarge and Large versions of EmoNeXt exhibit outstanding performance, both achieving a perfect accuracy score of 100. This matches the best current state-of-the-art model, EfficientNet-XGBoost (100), and surpasses several other models: ViT-SE by 0.2 (99.8), Inception-V3 fine-tuning by 0.43 (99.57), the Triple-Structure network by 0.96 (99.04), and MBCC-CNN by 1.52 (98.48). These

Table 3 Performance comparison on FER2013 test set

Model	Accuracy (%)
GoogleNet [58]	65.20
Deep Emotion [59]	70.02
Inception [60]	71.60
Ad-Corre [28]	72.03
SE-Net50 [61]	72.50
ResNet50 [61]	73.20
VGG [60]	73.28
EmoNeXt-Tiny	73.34
Residual Masking Network [62]	74.14
LHC-NetC [25]	74.28
EmoNeXt-Small	74.33
LHC-Net [25]	74.42
EmoNeXt-Base	74.91
CNNs + BOVW [63]	75.42
EmoNeXt-Large	75.57
Segmentation VGG-19 [31]	75.97
EmoNeXt-XLarge	76.12

results underscore the powerful generalization capability of EmoNeXt-XLarge and EmoNeXt-Large.

The table also highlights the performance of EmoNeXt’s Tiny, Small, and Base versions. Although EmoNeXt-XLarge and EmoNeXt-Large achieved the highest accuracy, the other versions remain competitive. For instance, EmoNeXt-Tiny attained an accuracy of 95.1, outperforming models such as the Multi Attention network (94.51) and Hybrid CNN (94.82).

EmoNeXt-Small demonstrated improved performance with an accuracy of 97.78, surpassing advanced architectures like the WMCNN and LSTM combination (97.5). EmoNeXt-Base also maintained strong performance, achieving an accuracy of 98.44, which exceeds the performance of the Dynamic Cascaded Classifier (97.84) and DeepEmotion (98).

Results on the AffectNet-7 dataset.

In Table 5, we present the results of our EmoNeXt models on the AffectNet-7 dataset. The XLarge version of EmoNeXt achieves state-of-the-art performance on AffectNet-7. Specifically, compared to DDAMFN (67.03) and POSTER (67.31), EmoNeXt-XLarge shows improvements of 0.43 and 0.15, respectively. Additionally, EmoNeXt-XLarge surpasses CF-DAN (63.58) by 0.55. The other versions of EmoNeXt also exhibit competitive performance, with EmoNeXt-Tiny achieving an accuracy of 65.55, outperforming models such as PAENet (65.29), DACL (65.20), MVT (64.57), and DLN (63.7). Additionally, EmoNeXt-Base and EmoNeXt-Small demonstrates improved performance with accuracies of 66.21 and 65.9, respectively. These results surpass advanced architectures like Multi-task EfficientNet-B0 (65.74) and DAN (65.69). EmoNeXt-Large also maintained strong performance, achieving an accuracy of 66.88, which exceeds the performance of EmotionGCN (66.46) and TransFER (66.23).

Table 4 Performance comparison on CK+ test set

Model	Accuracy (%)
LBP-CNN [64]	79.5
Broad learning [65]	81
STF + LSTM [66]	82
ResNet150 [67]	89
LTP-CNN [64]	89.2
CLBP-CNN [64]	91
LBP+ORB features [68]	93.2
Deep Features + HOG [69]	94.17
Multiple attention network [70]	94.51
hybrid CNN model [71]	94.82
EmoNeXt-Tiny	95.1
Pre-trained VGG19 [72]	96
WMCNN-LSTM [73]	97.5
Pre-trained Resnet50 [72]	97.7
EmoNeXt-Small	97.78
Dynamic cascaded classifier [74]	97.84
DeepEmotion [59]	98
EmoNeXt-Base	98.44
MBCC-CNN [75]	98.48
Pre-trained MobileNet [72]	98.5
Tripe-Structure Network [26]	99.04
Inception-V3-based approach [34]	99.57
ViT + SE [76]	99.8
EfficientNet-XGBoost [37]	100
EmoNeXt-Large	100
EmoNeXt-XLarge	100

Table 5 Performance comparison on AffectNet-7 ways validation set

Model	Accuracy (%)
EfficientFace [77]	63.70
CNNs and BOVW + local SVM [63]	63.31
CNNs and BOVW + global SVM [63]	62.94
Ad-Corre [28]	63.36
DLN [50]	63.7
MVT [52]	64.57
DACL [24]	65.20
PAENet [51]	65.29
EmoNeXt-Tiny	65.55
DAN [38]	65.69
Multi-task EfficientNet-B0 [78]	65.74
EmoNeXt-Small	65.9
EmoNeXt-Base	66.21
TransFER [79]	66.23
EmotionGCN [80]	66.46
EmoNeXt-Large	66.88
DDAMFN [36]	67.03
POSTER [41]	67.31
EmoNeXt-XLarge	67.46

Results on the AffectNet-8 dataset.

In Table 6, we present the results of our EmoNeXt models on the AffectNet-8 dataset. We observe that EmoNeXt-XLarge closely matches the best scores obtained by current state-of-the-art model on AffectNet-8 with a difference of 0.12 with DDAMFN (64.25). When compared to POSTER (63.34) and CF-DAN (63.58), EmoNeXt-XLarge improves by 0.15 and 0.55, respectively. The other versions of EmoNeXt also exhibit competitive performance. For instance, EmoNeXt-Tiny achieves an accuracy of 61.36, surpassing models like EfficientFace (59.89). Similarly, EmoNeXt-Base and EmoNeXt-Small demonstrate improved performance, achieving accuracies of 62.94 and 62.51, respectively. These results surpass advanced architectures like Multi-task EfficientNet-B0 (61.32) and DAN (62.09). EmoNeXt-Large also maintains strong performance, achieving an accuracy of 66.88, which exceeds the performance of EmotionGCN (66.46) and TransFER (66.23).

Overall, these results highlight the robustness and versatility of the EmoNeXt model across datasets with varying complexity and numbers of emotion classes. The superior performance of EmoNeXt-XLarge on AffectNet-7 and its competitive performance on AffectNet-8 demonstrate its effectiveness. Additionally, the strong results obtained by the other versions of EmoNeXt, including Tiny, Small, Base, and Large, indicate that the model scales well across different sizes, maintaining high accuracy and outperforming many existing state-of-the-art models. These consistent results underscore the strength of the EmoNeXt architecture in facial expression recognition tasks. Furthermore, the outstanding performance on AffectNet, a large-scale dataset with a substantial amount of data, underscores the model’s ability to handle and excel with large, complex datasets, making it a robust choice for real-world applications involving extensive data.

Table 6 Performance comparison on AffectNet-8 ways validation set

Model	Accuracy (%)
CNNs and BOVW + local SVM [63]	59.58
CNNs and BOVW + global SVM [63]	59.30
EfficientFace [77]	59.89
EmoNeXt-Tiny	61.36
MVT [52]	61.4
Multi-task EfficientNet-B0 [78]	61.32
DAN [38]	62.09
EmoNeXt-Small	62.51
EmoNeXt-Base	62.94
EmoNeXt-Large	63.12
POSTER [41]	63.34
CF-DAN [42]	63.58
EmoNeXt-XLarge	64.13
DDAMFN [36]	64.25

4.3.3 Application to detecting emotional responses in elderly individuals with Alzheimer’s disease

To assess the real-world applicability of our EmoNeXt model, we conducted experiments using publicly available interview videos featuring patients diagnosed with Alzheimer’s Disease (AD). These videos, sourced from various online platforms, showcase spontaneous facial expressions exhibited during interactions. Due to the absence of ground truth annotations for these videos, our evaluation primarily focuses on the qualitative performance of EmoNeXt in a real-time inference scenario.

The EmoNeXt model, trained on the FER2013 dataset, was employed to process the video frames and predict the emotional states of the patients. Throughout the interviews, the model consistently detected a predominance of negative emotions, such as fear, sad and anger, along with neutral expressions, and occasional instances of happiness. Specifically, the analyzed frames revealed emotions including fear, anger, sad, neutral, and happiness. These observations resonate with the emotional challenges commonly encountered by AD patients [81]. Remarkably, despite being infrequent, the model accurately identified moments where the elderly individual expressed happiness through smiles.

Figure 6 illustrates examples of frames from an interview video, highlighting the predicted emotions by EmoNeXt. For privacy reasons, the faces in the presented frames have been blurred. This measure safeguards the anonymity of individuals while enabling the demonstration of EmoNeXt’s precise emotion detection and classification capabilities, thereby showcasing its robustness in processing real-world data and validating its applicability beyond controlled datasets. In the frame on the top left, the elderly individual with AD expresses a negative emotion of fear, detected at 40.60% among all other emotions on which the model was trained. However, no element stimulating fear was identified in this part of the interview. In the frame on the top right, another negative emotion, anger, is detected with a score of 36.59%, despite the absence of any discussion element that could provoke anger in the elderly person. In the bottom left frame, a lack of emotion or neutral state is dominant, an emotion that prevailed throughout the interview alongside negative emotions. The fourth example, illustrated in the bottom right frame, shows an emotion of joy detected at 79.60%. This emotion was rarely identified during the interview but was noted during joyful parts of the discussion.

These findings are consistent with studies in neuropsychiatry and neuroscience, which indicate that individuals with cognitive impairments, such as those with AD, often express fewer positive emotions and more negative emotions compared to cognitively unimpaired individuals [20–23]. They also frequently have difficulty recognizing and processing facial expressions, which can serve as early indicators for seeking diagnosis or treatment.

These insights emphasize the relevance of EmoNeXt in providing a nuanced understanding of the emotional expressions of AD patients, contributing to more effective and empathetic care strategies. By accurately identifying and classifying subtle emotional cues, EmoNeXt can aid caregivers and clinicians in monitoring the emotional well-being of AD patients, ultimately enhancing the quality of care provided.

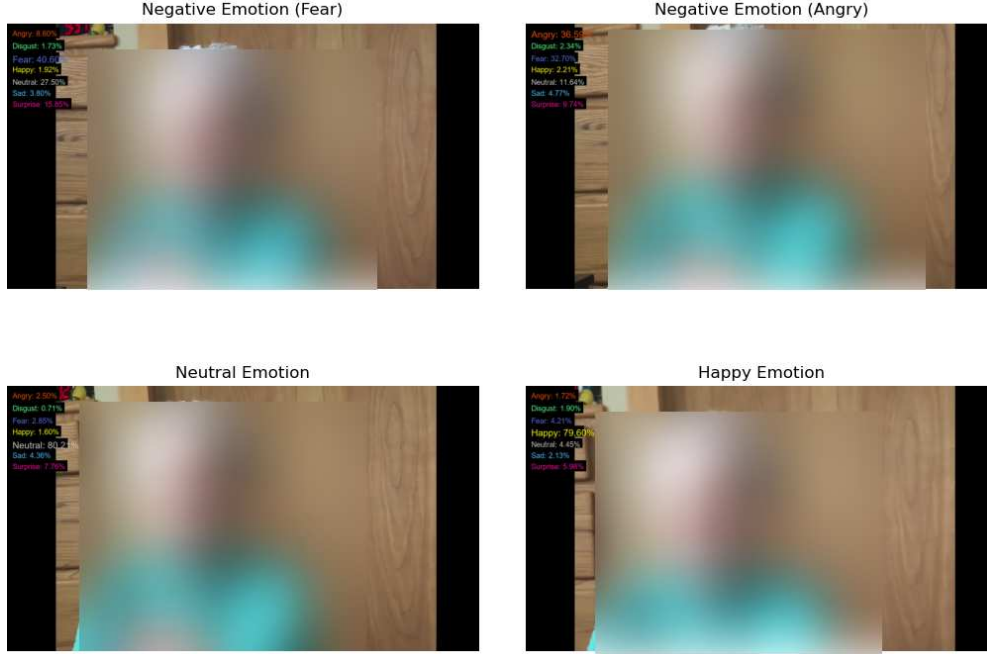


Fig. 6 Examples of emotion detection in an interview video of an Alzheimer’s Disease patient using EmoNeXt. The predicted emotions are overlaid on the video frames. Faces have been blurred for privacy reasons. Source: [82].

5 Conclusion

In this study, we have expanded the evaluation of our deep learning framework for facial emotion recognition, EmoNeXt, originally introduced in prior work. EmoNeXt was initially tested using the FER2013 dataset, where it demonstrated superior performance compared to existing models. In this paper, we extended the evaluation to include the AffectNet and CK+ datasets, two prominent and diverse collections of facial expression data. This broader assessment revealed that EmoNeXt consistently outperforms both the original ConvNeXt network and other state-of-the-art FER models across multiple datasets. Our comprehensive evaluation included an extensive ablation study to understand the impact of each component integrated into EmoNeXt. This analysis not only highlighted the contributions of individual modules but also examined their collective effects on emotion recognition performance, providing deeper insights into the mechanisms driving EmoNeXt’s success.

Additionally, we explored the practical application of EmoNeXt in the healthcare sector, particularly for elderly individuals with Alzheimer’s disease (AD). Given the communication challenges faced by AD patients due to neurodegenerative effects, accurate emotion detection is crucial for enhancing their care. Our study demonstrated EmoNeXt’s ability to detect emotions such as neutrality, anger, sadness, and joy in

AD patients, underscoring its potential to improve their quality of life. These findings emphasize the relevance of EmoNeXt in providing nuanced understanding and monitoring of emotional expressions in AD patients, offering valuable insights for caregivers and clinicians. By accurately identifying and classifying subtle emotional cues, EmoNeXt can contribute to more effective and empathetic care strategies.

6 Future work

As part of our ongoing research, we plan to conduct clinical trials involving elderly residents in French nursing homes, particularly those diagnosed with neurodegenerative diseases such as Alzheimer’s and Parkinson’s. This collaboration with our healthcare partners will enable us to test EmoNeXt in a controlled clinical setting. Additionally, one of our future goals is to deploy the EmoNeXt model in companion robots and virtual agent applications to facilitate patient-machine interactions, thereby enhancing the support provided to AD patients. By integrating EmoNeXt into these technologies, we aim to improve the emotional support and monitoring capabilities available to caregivers and clinicians, ultimately contributing to more effective and empathetic care strategies.

However, it is important to note the limitations of relying solely on facial expressions for emotion detection, especially in the context of AD, where verbal and non-verbal communication may be impaired. Emotions are complex and multifaceted, often requiring the integration of multiple modalities for more accurate and comprehensive recognition. Therefore, future work will focus on incorporating additional modalities such as voice, text, gestures, and biosignals to enhance the performance and reliability of emotion recognition systems. To support this direction, we plan to utilize and further develop our previously introduced multimodal emotion recognition dataset [83], which integrates facial expressions, voice and gestures. This dataset will serve as a valuable resource for evaluating and improving multimodal emotion recognition models.

Declarations

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6.2 Competing Interests

The authors have no relevant financial or non-financial interests to disclose. There are no competing interests to declare that are relevant to the content of this article. All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript. Furthermore, the authors have no financial or proprietary interests in any material discussed in this article.

6.3 Availability of Data and Materials

The code for the model and experiments is available at <https://github.com/yelboudouri/EmoNeXt>.

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